SOMAIYA VIDYAVIHAR UNIVERSITY

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<tr><td>Maximum Marks: 30</td><td colspan="2">Semester: Aug 2025 – Nov 2025
Examination: In-Semester Examination
MSE</td><td>Duration: 1 Hr 15 Min</td></tr>
<tr><td colspan="2">Programme code:
Programme: B.Tech</td><td>Class: FY</td><td>Semester: I (SVU R-2025)</td></tr>
<tr><td colspan="2">Name of the Constituent College:
K. J. Somaiya College of Engineering</td><td colspan="2">Name of the department:
COMP/ETRX/EXTC/IT/MECH</td></tr>
<tr><td>Course Code: 216U06C101</td><td colspan="3">Name of the Course: Applied Mathematics-I</td></tr>
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<tr><th>Question No.</th><th></th><th>Max. Marks</th></tr>
<tr><td>Q:1</td><td>Find Eigen values & eigen vectors of $A^{50}+2A^{25}+I$ & Adj A if $A=\begin{bmatrix}0&1&0\\0&0&1\\1&-3&3\end{bmatrix}$
Solution:
Characteristic equation: $|A-\lambda I|=0$
$\therefore \begin{vmatrix}-\lambda&1&0\\0&-\lambda&1\\1&-3&3-\lambda\end{vmatrix}=0$
$\therefore \lambda^3-3\lambda^2+3\lambda+1=0$
$\therefore \lambda=1,1,1$ are the eigenvalues of A.

For $\lambda=1 [A-I]X=0$
$\therefore \begin{bmatrix}-1&1&0\\0&-1&1\\1&-3&2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=0$
R3+R1⇒ $\begin{bmatrix}-1&1&0\\0&-1&1\\0&-2&2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=0$
R3-2R2 ⇒ $\begin{bmatrix}-1&1&0\\0&-1&1\\0&0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=0$
$\therefore x_2=x_3$ &
$\therefore x_1=x_2$
$\therefore$ Eigen vector for matrix A corresponding to $\lambda=1$ is $\begin{bmatrix}1\\1\\1\end{bmatrix}$

Eigen values for matrix $A^{50}+2A^{25}+$ I are : $\lambda^{50}+2\lambda^{25}+1$:
i.e. $1+2(1)+1=4,4,4$ with corresponding eigen vector $\begin{bmatrix}1\\1\\1\end{bmatrix}$
Eigen values for matrix AdjA are: $\frac{|A|}{\lambda}$.
Here $|A|=$ product of eigen values of A $=1$
Eigen values for matrix AdjA are 1,1, 1 with corresponding eigen vector $\begin{bmatrix}1\\1\\1\end{bmatrix}$</td><td>06

…….03

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<tr><td>Q:2</td><td>Attempt any TWO from the following.</td><td>12</td></tr>
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1	If A $= \begin{bmatrix} -1 & 4 \\ 2 & 1 \end{bmatrix}$ then prove that 3 tan A= A tan 3. Solution: The characteristic equation of A is $\begin{vmatrix} -1-\lambda & 4 \\ 2 & 1-\lambda \end{vmatrix} = 0$ $\therefore \quad (1+\lambda)(1-\lambda) - 8 = 0 \qquad \therefore \lambda^2 - 9 = 0$ $\therefore \quad \lambda = 3, -3.$ **(i)** For $\lambda = 3$, $[A - \lambda I]\, X = 0$ gives $\begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ By $\quad R_2 + \frac{1}{2}R_1 \begin{bmatrix} -4 & 4 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\therefore \quad -4x_1 + 4x_2 = 0 \qquad \therefore \quad x_1 - x_2 = 0$ Putting $x_2 = t$, we get $x_1 = t$. $x_1 = \begin{bmatrix} t \\ t \end{bmatrix} = t\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \therefore \quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ The eigen vector is $[1,\ 1]'$. For $\lambda = -3$, $[A - \lambda I]\, X = 0$ gives $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ By $\quad R_2 - R_1 \begin{bmatrix} 2 & 4 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\therefore \quad 2x_1 + 4x_2 = 0 \qquad \therefore \quad x_1 + 2x_2 = 0$ Putting $x_2 = -t$, we get $x_1 = -2x_2 = 2t$. $\therefore \quad x_2 = \begin{bmatrix} 2t \\ -t \end{bmatrix} = t\begin{bmatrix} 2 \\ -1 \end{bmatrix} \qquad \therefore x_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ $\therefore$ The eigen vector is $[2,\ -1]'$ $\therefore \quad M = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$ and $\lvert M \rvert = -3$ $M^{-1} = \dfrac{adj.M}{\lvert M \rvert} = -\dfrac{1}{3}\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$ Now D $= \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix}$ $\because \ f(A) = \tan A$ $f(D) = \begin{bmatrix} \tan 3 & 0 \\ 0 & \tan(-3) \end{bmatrix} = \begin{bmatrix} \tan 3 & 0 \\ 0 & -\tan 3 \end{bmatrix}$ $\therefore \tan A = M f(D) M^{-1}$ $= \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} \tan 3 & 0 \\ 0 & \tan(-3) \end{bmatrix}\left(-\frac{1}{3}\right)\begin{bmatrix} -1 & -2 \\ -1 & 1 \end{bmatrix}$ $= -\frac{1}{3}\begin{bmatrix} \tan 3 & -2\tan 3 \\ \tan 3 & \tan 3 \end{bmatrix}\begin{bmatrix} -1 & -2 \\ -1 & 1 \end{bmatrix}$ $= -\frac{1}{3}\begin{bmatrix} \tan 3 & -4\tan 3 \\ -2\tan 3 & -\tan 3 \end{bmatrix}$ $\therefore \quad 3\tan A = \begin{bmatrix} -\tan 3 & 4\tan 3 \\ 2\tan 3 & \tan 3 \end{bmatrix} = \tan 3 \begin{bmatrix} -1 & 4 \\ 2 & 1 \end{bmatrix}$	06 ……..03 ……..06
2	If $y = tan^{-1}x$ then Prove that $y_n = (-1)^{n-1}(n-1)!\, sin^n\theta \sin n\theta$ where $\theta = tan^{-1}\left(\frac{1}{x}\right)$	06

❖ **Solution:** Differentiating $y = \tan^{-1} x$, w.r.t. x, we get,

$$y_1 = \frac{1}{x^2+1} = \frac{1}{(x+i)(x-i)} = \frac{1}{2i}\left[\frac{1}{(x-i)} - \frac{1}{(x+i)}\right]$$ ……03

Differentiating $(n-1)$ times,

$$y_n = \frac{1}{2i}\left[\frac{(-1)^{n-1}(n-1)!}{(x-i)^n} - \frac{(-1)^{n-1}(n-1)!}{(x+i)^n}\right] \quad \text{.............(1)}$$

Putting $r = r\cos\theta$, $1 = r\sin\theta$ $r = \sqrt{1+x^2}$ and $\theta = \tan^{-1}(1/x)$,

$$\frac{1}{(x-i)^n} = \frac{1}{r^n(\cos\theta - i\sin\theta)^n} = \frac{1}{r^n}(\cos n\theta + i\sin n\theta)$$

Similarly, $$\frac{1}{(x+i)^n} = \frac{1}{r^n(\cos\theta + i\sin\theta)^n} = \frac{1}{r^n}(\cos n\theta - i\sin n\theta)$$

$$\therefore \quad \frac{1}{(x-i)^n} - \frac{1}{(x+i)^n} = \frac{2i}{r^n}\sin n\theta$$ …….06

from (1), we get,

$$y_n = \frac{(-1)^{n-1}(n-1)!}{2i} \cdot \frac{2i}{r^n}\sin n\theta$$
$$= (-1)^{n-1}(n-1)! \cdot \frac{1}{r^n}\sin n\theta \quad \ldots$$

3 Using Taylor's theorem, arrange in powers of x. 06

$-(x+2)^5 + (x+2)^4 + 3(x+2)^3 + (x+2) + 7$

Solution: Given function can be written as $f(x+2)$ and can be compared with $f(x+h)$.

Here, $f(x) = -x^5 + x^4 + 3x^3 + x + 7$ & $h = 2$ and as expansion required in powers of x, we can use

$f(x+h) = f(h) + xf'(h) + \frac{x^2}{2!}f''(h) + \cdots$….(1)

$f(x) = -x^5 + x^4 + 3x^3 + x + 7$ $\qquad f(2) = 17$

$f'(x) = -5x^4 + 4x^3 + 9x^2 + 1$ $\qquad f'(2) = -11$

$f''(x) = -20x^3 + 12x^2 + 18x$ $\qquad f''(2) = -76$

$f'''(x) = -60x^2 + 24x + 18$ $\qquad f'''(2) = -174$ …….03

$f^{iv}(x) = -120x + 24$ $\qquad f^{iv}(2) = -216$

$f^{v}(x) = -120$ & $f^{vi}(x) = 0$ $\qquad f^{v}(2) = -120$

Putting these values in (1) we get,

$-(x+2)^5 + (x+2)^4 + 3(x+2)^3 + (x+2) + 7$

$= f(2) + xf'(2) + \frac{x^2}{2!}f''(2) + \frac{x^3}{3!}f'''(2) + \frac{x^4}{4!}f^{iv}(2) + \frac{x^5}{5!}f^{v}(2)$

$= 17 - 11x - 38x^2 - 29x^3 - 9x^4 - x^5$ ……..06

Q:3 Attempt any TWO from the following. 12

1 Verify Cayley-Hamilton theorem for matrix A, hence find 06

A^{-1} & A^4. $A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$

Soln. The characteristic equation is given by $|A - \lambda I| = 0$

$-\lambda^3 + S_1\lambda^2 - S_2\lambda + |A| = 0$

Where S_1=Trace A

S_2= Sum of Minors of Diagonal Elements

$|A|$ = Determinant of A

For given Matrix A,

S_1=6, S_2=11, $|A| = 6$

$\therefore f(\lambda) = \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$: The characteristic equation

By Cayley-Hamilton Theorem, A Should satisfy the characteristic equation. …….02

Verification:

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 0 & -4 \\ 0 & 4 & 0 \\ -4 & 0 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 14 & 0 & -13 \\ 0 & 8 & 0 \\ -13 & 0 & 14 \end{bmatrix}$$

$$\therefore f(A) = A^3 - 6A^2 + 11A - 6I$$

$$= \begin{bmatrix} 14 & 0 & -13 \\ 0 & 8 & 0 \\ -13 & 0 & 14 \end{bmatrix} - 6\begin{bmatrix} 5 & 0 & -4 \\ 0 & 4 & 0 \\ -4 & 0 & 5 \end{bmatrix} + 11\begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix} - 6I$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence , $\therefore f(A) = A^3 - 6A^2 + 11A - 6I = 0$……….(1) ……..04

Ie. A satisfies its characteristic equation. Hence Cayley-Hamilton theorem is verified.

To find A^{-1},

Pre-multiplying (1) by A^{-1}

$$\therefore A^2 - 6A + 11I - 6\,A^{-1} = 0$$

$$\therefore\ A^{-1} = \frac{1}{6}\begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$$

To find A^4,

Pre-multiplying (1) by A

$$\therefore A^4 - 6A^3 + 11A^2 - 6A = 0$$

$$\therefore\ A^4 = \begin{bmatrix} 41 & 0 & -40 \\ 0 & 16 & 0 \\ -40 & 0 & 41 \end{bmatrix}$$

……..06

2

Prove That $\int_0^a x^4\,(a^2 - x^2)^{\frac{1}{2}}\,dx = \frac{a^6\pi}{32}$ 06

❖ **Solution:**

Put $x^2 = a^2 t$

$x = a\,t^{\frac{1}{2}}, \quad dx = \frac{a}{2}\,t^{\frac{1}{2}-1}\,dt$

$$I = \int_{t=0}^{1} a^4 t^2\,(a^2 - a^2 t)^{\frac{1}{2}}\,\frac{a}{2} t^{\frac{1}{2}-1}\,dt$$

$$= a^4 \times a \cdot \frac{a}{2}\int_{t=0}^{1} t^2\,(1-t)^{\frac{1}{2}}\,t^{\frac{1}{2}-1}\ dt$$

$$= \frac{a^6}{2}\int_{t=0}^{1} t^{\frac{5}{2}-1}\,(1-t)^{\frac{3}{2}-1}\ dt$$ ……..03

$$= \frac{a^6}{2}\,\beta\left(\frac{5}{2},\ \frac{3}{2}\right)$$

	$= \frac{a^6}{2} \frac{\overline{\left|\frac{5}{2}\right.}\overline{\left|\frac{3}{2}\right.}}{\overline{\left|4\right.}}$ $= \frac{a^6}{2} \times \frac{\frac{3}{2}\times\frac{1}{2}\sqrt{\pi}\times\frac{1}{2}\times\sqrt{\pi}}{3.2.1} = \frac{a^6\pi}{32}$	……06
3	Using DUIS, evaluate, $\int_0^\infty \frac{e^{-x}}{x}(1-e^{-ax})\,dx, \quad a>-1$ **Solution:** Let, $I(a) = \int_0^\infty \frac{e^{-x}}{x}(1-e^{-ax})\,dx \dots\dots\dots\dots\dots\dots (1)$ Using differentiation under Integral Sign w. r. t. ‘a’ $\frac{dI}{da} = \int_0^\infty \frac{\partial}{\partial a}\frac{e^{-x}}{x}(1-e^{-ax})\,dx$ $= \int_0^\infty \frac{e^{-x}}{x}\big(0-e^{-ax}(-x)\big)\,dx$ $= \int_0^\infty e^{-(1+a)x}\,dx = \left.\frac{e^{-(1+a)x}}{-(1+a)}\right\|_0^\infty$ $= -\frac{1}{1+a}[e^{-\infty}-e^0]$ $= -\frac{1}{1+a}[0-1]$ $\therefore \frac{dI}{da} = \frac{1}{1+a}$ Integrating both sides w.r.to a we get, $I(a) = \log(1+a) + c \dots\dots\dots\dots\dots (2)$ To find c put $a=0$ we get, $I(0) = \log 1 + c$ But, from (1) $I(0) = 0$ $\therefore c = 0$ $\therefore$ From (2) $I(\alpha) = \int_0^\infty \frac{e^{-x}}{x}(1-e^{-ax})\,dx = \log(1+a)$	06 ……..03 …….06